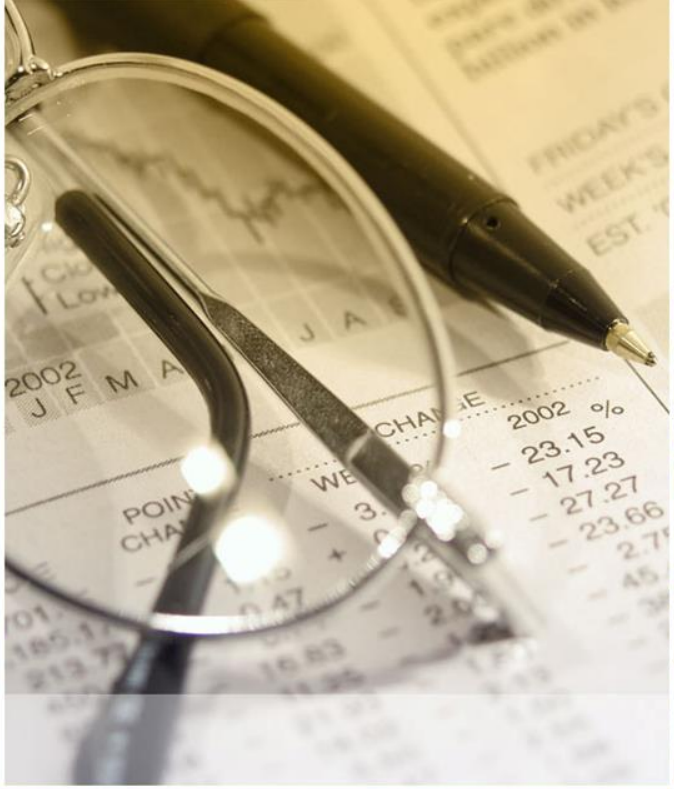




Capital Cost Overruns in the Mining Industry:

Mistakes, Misjudgments or ?

June 14, 2015

Presented by:

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TETRA TECH





Objectives of the Presentation

- **Identify** the contributing factors to capital cost overruns in the mining industry; and
- **Quantify** the relevance of the each contributing factor



Increases in Capital & Operating Costs and IRR

Rising Costs	
Capital Costs	Operating Costs
+44%	+20%

Average Return
IRR[1]
+20%

[1] *Based on mean after-tax calculations only.*

Source: Silver, Doug B. "How to Avoid Rising Capital and Operating Costs", Denver Gold Show, September 8, 2008.





Contingency And Accuracy:

Capital Cost: Mistakes or Misjudgments?

Description	AACE Designation[1]	AACE: Accuracy Ranges[1]	AACE: % Completion of Project Definition Deliverables[1]	Contingency [range/ average][2]	Accuracy [range/ average][2]
Scoping or Preliminary Assessment	Class 5	Low: -20% to -50% High: +30% to +100%	0% - 2%	20-35%/24%	30-50%/40%
Preliminary Feasibility Study	Class 4	Low: -15% to -30% High: +20% to +50%	1% - 15%	10-20%/17%	20-30%/20%
Feasibility Study	Class 3	Low: -10% to -20% High: +10% to +30%	10% - 40%	5-20%/13%	12-20%/16%
Basic Engineering	Class 2	Low: -5% to -15% High: +5% to +20%	30% - 75%	4-15%/9%	3-8%/5.5%

[1] *AACE International Recommended Practice No. 47R-1, "Cost Estimate Applied in the Mining and Mineral Processing Industries", Rev. July 6, 2012*

[2] *Bullock, Richard L. "Accuracy of feasibility study evaluations would improve accountability", Jackling Lecture, Mining Engineering, April 2011*





Key Areas of Concern

Area of Problem	Frequency
Geology, resource and reserve estimation	**17%**
Geotechnical analysis	**9%**
Mine Design and Scheduling	**32%**
Mining Equipment Selection	**4%**
Metallurgical Test Work, Sampling and Scale-Up	**15%**
Process Plant Equipment Design and Installation	**12%**
Cost Estimation	**7%**
Hydrology	**4%**

Source: McCarthy, Peter. "Why Feasibility Studies FAIL", AusIMM Melbourne Branch, 2013.



Sources for Survey (80% Response Rate)

Disciplines Surveyed

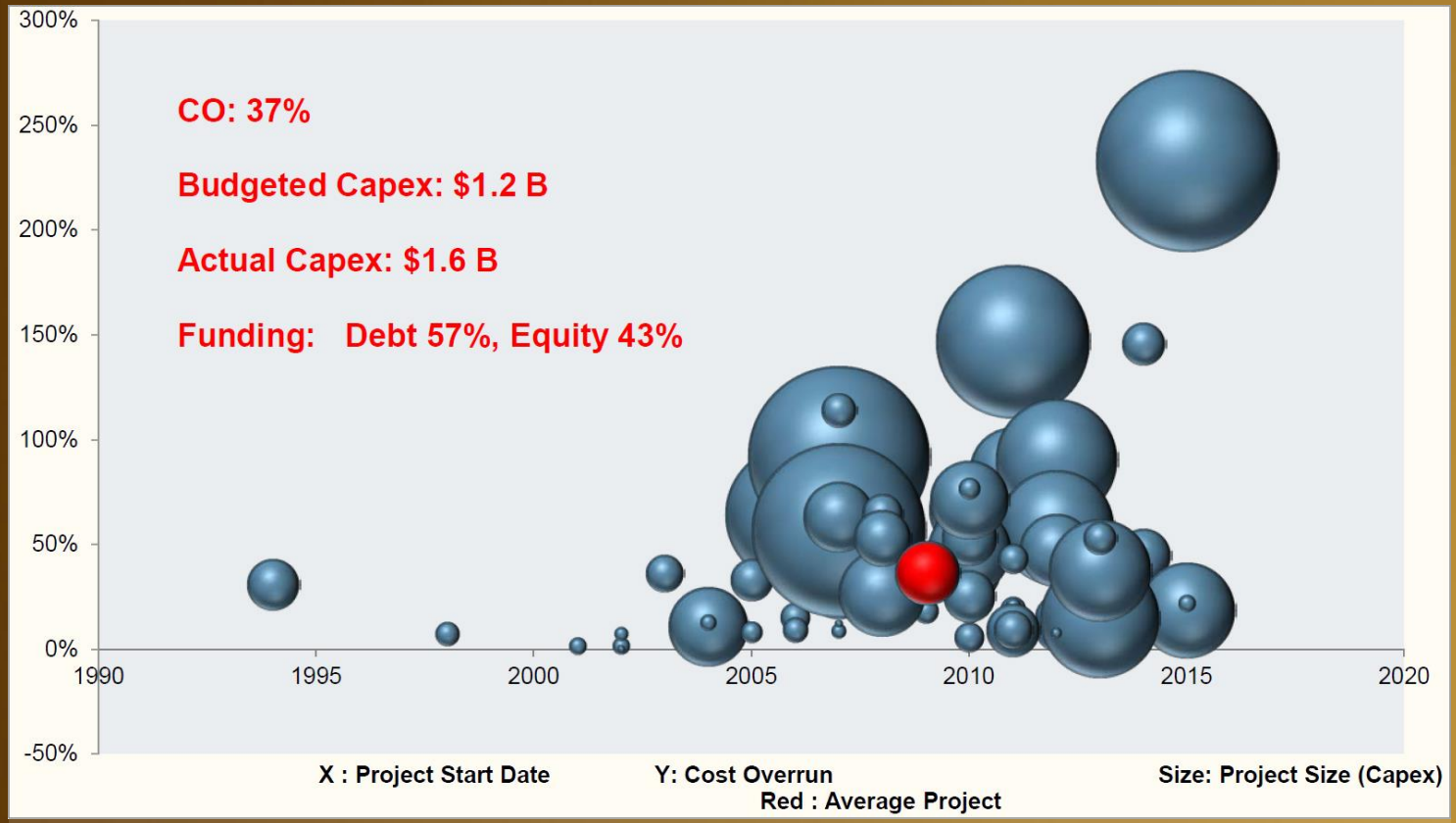
- Geologists
- Mining Engineers
- Process Metallurgical Engineers
- Mineral Economists
- Mining Management
- Financial Community
- Educators

Areas Surveyed

- Australia
- Brazil
- Canada
- Chile
- Dominican Republic
- Europe
- Mexico
- South Africa
- United States



Average Mining Project



Source: Lwin, Tin, and Jose Lazo. 'Project Cost Overruns - A Lender's Perspective'. 2015. Presentation.



Studies of Mine Performances

McCarthy and Ward (1999)	Start-up performance of 9 Australian underground base metal mines showed that only 50% achieved design throughput by Year 3 and 25% never achieved it at all.
McCarthy (2004)	Results of 15 year study of 56 mines with Feasibility Studies showed that only 20% were successful in achieving the design production capacity.
Tatman (2003)	Study of 41 underground mines showed that 60% of ore reserves were outside of expected range, some very seriously in error.
Tatman (2004)	Study showed that Final Feasibility Study production rates of 60 steeply-dipping tabular deposits did not achieve planned production.
Bertisen and Davis (2008)	Study of 63 mines and smelters worldwide completed between 1980 and 2001 showed that Feasibility Study capital cost estimates averaged overruns of 25% on an as-built in actual (nominal) dollars and 14% after as-built costs when adjusted for inflation.
Bullock (2011)	Summary of 8 studies of mining projects from 1965 to 2002 showed a weighted average capital cost overrun of +25% with 58 to 73 projects having cost overruns of greater than 15% but less than 100%, and 8 with capital cost overruns greater than 100%.





Survey Questions

- Aggressive and unrealistic schedules
- Costs escalated and/or factored from historic studies
- Lack of properly developed testwork/defined design criteria/work scope
- Inflation and/or change in exchange rates
- Poor estimating methods (missing data/information)
- Over-estimated accuracy/under-estimated contingency
- Client pressure to minimize the initial capital requirements
- Client pressure to quickly complete work
- Selection of wrong mining method and/or processing flow sheet

Survey Questions (Continued)

- Under estimation of transportation and logistics
- Under estimation and/or insufficient critical data from early engineering studies
- Unrepresentative samples for geotechnical, environmental and/or metallurgical testing
- Unexpected social, cultural, political and/or environmental issues
- Estimates based on incomplete/insufficient bid packages
- Owner and/or Engineer that is not experienced in pragmatic delivery of projects
- Major changes in the funding and/or ownership of projects



Additional Survey Comments

- Study team wanting/needing to deliver good news to company's management/board/investors (motive is more consulting/engineering work).
- Promises, often premature, made by company's study team to company's senior management/ board/investors.
- Jumping engineering/stages - i.e. going straight from PEA to FS.
- Poorly setup project controls.



Additional Survey Comments (Cont.)

- Discontinuity in project management and communications at different level of company's and engineering teams.
- Changing of project plan and implementation in mid-construction.
- Over-estimation of labor productivity/materials during construction (e.g. using North American productivities and material costs in foreign locations, but using foreign labor rates).





Results and Conclusions

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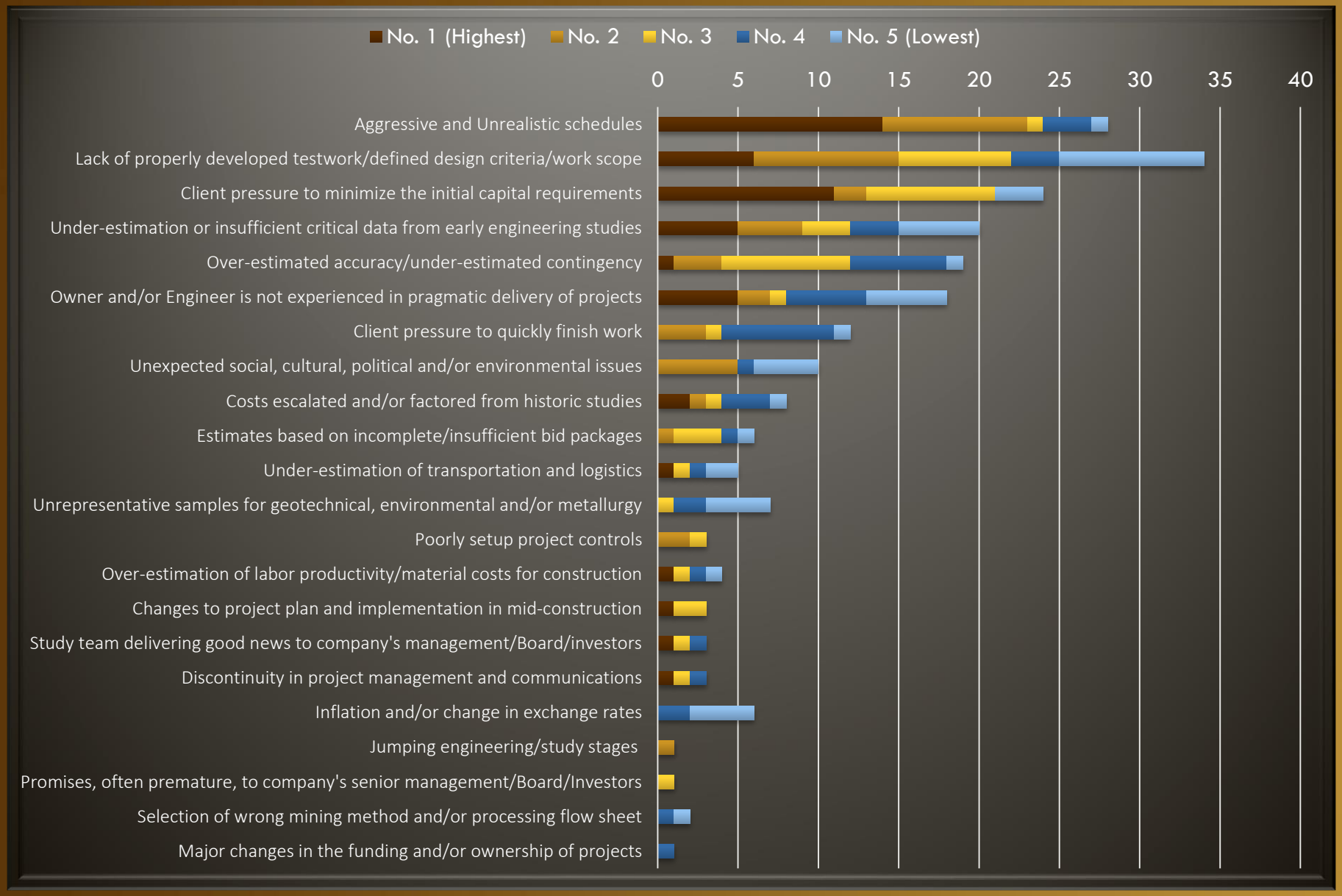
TOP 5 Contributing Factors

1. AGGRESSIVE AND UNREALISTIC SCHEDULES (116)
2. LACK OF PROPERLY DEVELOPED TESTWORK/ DEFINED DESIGN CRITERIA/WORK SCOPE (102)
3. CLIENT PRESSURE TO MINIMIZE THE INITIAL CAPITAL REQUIREMENTS (90)
4. UNDER-ESTIMATION OR INSUFFICIENT CRITICAL DATA FROM EARLY ENGINEERING STUDIES (61)
5. OVER-ESTIMATED ACCURACY/UNDER-ESTIMATED CONTINGENCY (54)

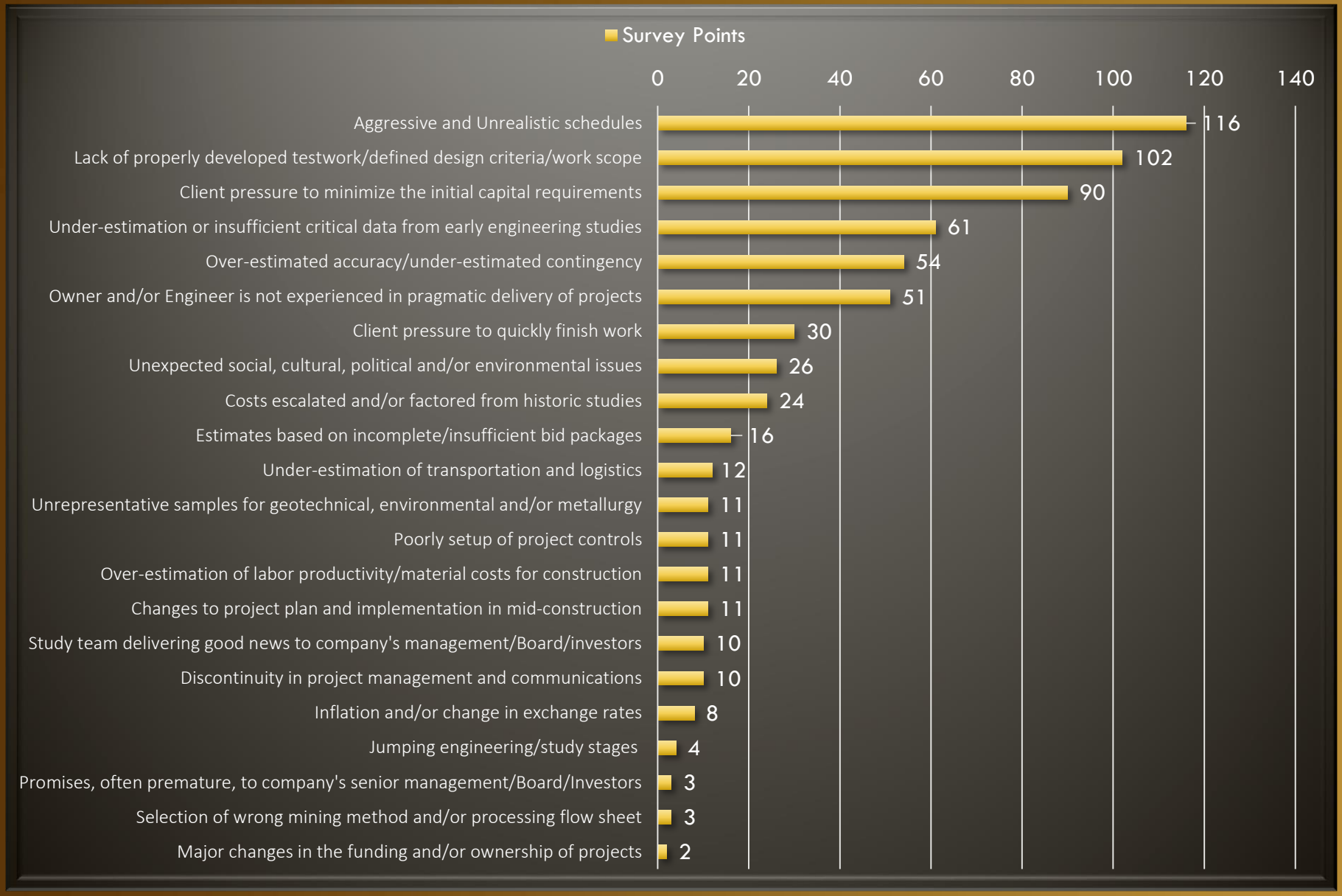


Description	Highest → Lowest					Survey Points
	No. 1	No. 2	No. 3	No. 4	No. 5	
Aggressive and Unrealistic schedules	14	9	1	3	1	116
Lack of properly developed testwork/defined design criteria/work scope	6	9	7	3	9	102
Client pressure to minimize the initial capital requirements	11	2	8	0	3	90
Under-estimation or insufficient critical data from early engineering studies	5	4	3	3	5	61
Over-estimated accuracy/under-estimated contingency	1	3	8	6	1	54
Owner and/or Engineer is not experienced in pragmatic delivery of projects	5	2	1	5	5	51
Client pressure to quickly finish work	0	3	1	7	1	30
Unexpected social, cultural, political and/or environmental issues	0	5	0	1	4	26
Costs escalated and/or factored from historic studies	2	1	1	3	1	24
Estimates based on incomplete/insufficient bid packages	0	1	3	1	1	16
Under-estimation of transportation and logistics	1	0	1	1	2	12
Unrepresentative samples for geotechnical, environmental and/or metallurgy	0	0	1	2	4	11
Poorly setup project controls	0	2	1	0	0	11
Over-estimation of labor productivity/material costs for construction	1	0	1	1	1	11
Changes to project plan and implementation in mid-construction	1	0	2	0	0	11
Study team delivering good news to company's management/Board/investors	1	0	1	1	0	10
Discontinuity in project management and communications	1	0	1	1	0	10
Inflation and/or change in exchange rates	0	0	0	2	4	8
Jumping engineering/study stages	0	1	0	0	0	4
Promises, often premature, to company's senior management/Board/Investors	0	0	1	0	0	3
Selection of wrong mining method and/or processing flow sheet	0	0	0	1	1	3
Major changes in the funding and/or ownership of projects	0	0	0	1	0	2

Responses per Question



Total Points per Question





The Seven Deadly Sins of Megaprojects

From Why Megaprojects fail so often, Edward Merrow, Industrial Megaprojects

I want to keep it all!

I want it now!

Don't worry, we'll work out the details later.

Why do we have to spend so much up front?

We need to shave 20 percent off that number!

The contractors should carry the risk, they're doing the project!

Fire *(or promote)* those #$@$^! Project Managers who overrun our projects!



Lessons Learned from the Investigation

- This topic has been examined from the mid-1960's until now… or 50 years.
- All these studies identified the same contributing factors to capital cost overruns as we are seeing now.
- In summary, the mining industry has not learned anything regarding capital cost overruns or made any corrections to its estimates.





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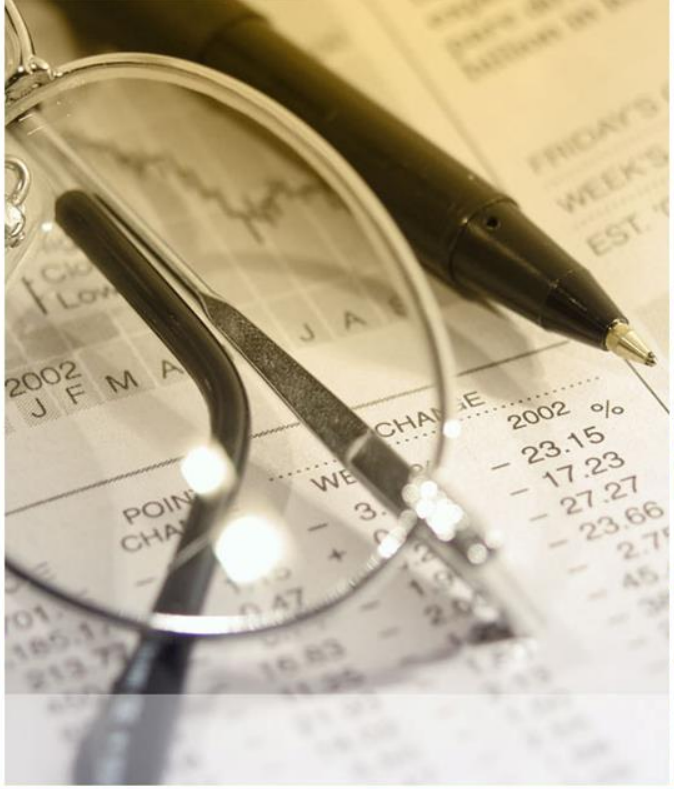
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Thanks

–Al Kuestermeyer,

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